Maximal Separated Phase Clocks and Spatial Charges for Cyclic Langton Ants

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Abstract

For a cyclic n-color Langton ant, we classify observables that separate the complete coloring from the head pose and have a constant increment on every update. The coloring statistic is allowed to be arbitrary, rather than assumed additive, local, or spatially periodic. In one invariant heading–checkerboard sector, the largest modulus of a unit-increment clock is $8n$ when the rule contains odd numbers of both right and left turns, and $4n$ otherwise. Explicit four-periodic spatial coefficients attain the bound. A coloring statistic shared by both sectors has maximal modulus $4n$. For the classical two-color ant this gives a sharp sixteen-phase clock, together with a complete classification of separated cocycles over every abelian coefficient group. For mixed rules we also classify all compactly supported coloring potentials: they are unique finite sums of explicit order-two plaquette charges, with the available face families determined by the two turn-count parities. The proofs use forced additivity, local pose equations, and planar integration. Exact Smith-form audits and independent literal implementations provide reproducible controls. The resulting formulas describe the exact phase information available from separated full-state observations.

1 Introduction

A cyclic Langton ant reads a cell, turns left or right according to its color, advances that color through a finite cycle, and moves one lattice step. The alternating horizontal and vertical moves partition its possible incoming poses into two checkerboard sectors. This representation, and its interpretation through local Truchet connections, go back to the study of generalized ants by Gale, Propp, Sutherland and Troubetzkoy [1].

A natural way to seek algebraic information about the dynamics is to combine a statistic of the entire coloring with a correction depending only on the position and heading. Such observables include familiar spatial charges and modular counters. We ask a completeness question: how much elapsed-time phase can any observable of this separated form retain, even if its coloring statistic is nonlinear and has unrestricted spatial dependence?

Our answer is exact. For a rule with r right-turn colors and l left-turn colors, $r + l = n$, a sector-specific unit clock has maximal modulus

$$M = \begin{cases} 8n, & r, l \text{ both odd}, \\ 4n, & \text{otherwise}. \end{cases} \tag{1}$$

The classical rule has $M = 16$. The extra factor of two requires spatial resolution beyond period two, and disappears when one insists on a single coloring statistic for both sectors. We additionally give a complete local presentation for the separated cocycles of every mixed rule, and a basis for all their compactly supported coloring potentials.

Relation to previous work. Jillhewar [2], in the proposition labelled `prop:cocycle`, proves the classical additive-charge conditions $4w = 0$ and the zero four-corner sum, with sufficiency over $\mathbb{Z}/4\mathbb{Z}$. We use and explicitly recover this zero-rate calculation; the familiar row and column charges are not newly introduced here. McNamara [3] studies fixed finite ansatz families with affine coordinate/time terms and zero-, one-, or two-bit site classes. That study already explains why grouping the increments by turn removes the order of the colors from the parameter equations. Our contribution is the unrestricted spatial and arbitrary-coloring problem, its sharp nonzero phase order, and the complete compact-charge basis. Order-blindness of the grouped parameter equations is credited to that earlier observation, not claimed anew.

2 All-state separated observables

Let $\tau = (\tau_0, \dots, \tau_{n-1}) \in \{\mathrm{R}, \mathrm{L}\}^n$ be nonempty. A coloring $C : \mathbb{Z}^2 \to \mathbb{Z}/n\mathbb{Z}$ belongs to $\mathcal{C}_n$ if it is zero outside a finite set. A state is $X = (C, p, h)$, where $p = (x, y)$ and $h \in \{0, 1, 2, 3\}$ denotes north, east, south, west, with step vectors $e_0 = (0, 1), e_1 = (1, 0), e_2 = (0, -1), e_3 = (-1, 0)$. Set $r = \#\{i : \tau_i = \mathrm{R}\}$ and $l = \#\{i : \tau_i = \mathrm{L}\}$, where $0 \le i < n$. Reading color $c = C(p)$ changes h by $+1$ for R or -1 for L, changes c to $c + 1 \pmod n$, and moves in the new heading. Denote this map by T_τ.

The value

$$\varepsilon = h - x - y \pmod 2$$

is conserved. We first fix $\varepsilon = 0$: even sites have incoming headings N/S and odd sites E/W. The quarter-turn $(x, y, h) \mapsto (-y, x, h - 1)$ carries the other sector to this one and preserves the rule. All statements quantify over every finite coloring and every pose in the stated sector, not merely over the orbit of a blank input.

Let G be an arbitrary abelian group and $\kappa \in G$. We call

$$I(C, p, h) = F(C) + \phi(p, h), \qquad I(T_\tau X) - I(X) = \kappa \tag{2}$$

a separated cocycle of constant rate κ. No initial assumption is made on F except that it is a function on $\mathcal{C}_n$. In particular it can initially be nonlinear or depend on the entire finite coloring. A conserved quantity has $\kappa = 0$. A unit phase clock modulo m has $G = \mathbb{Z}/m\mathbb{Z}$ and $\kappa = 1$.

Lemma 2.1 (Forced additivity)**.** *Every F in (2) has a unique representation*

$$F(C) = F(0) + \sum_{v \in \mathbb{Z}^2} f_v(c_v), \qquad f_v(0) = 0. \tag{3}$$

For a mixed rule, meaning $r, l > 0$, there exist $a_v, b_v \in G$ such that

$$f_v(c) = R(c)a_v + L(c)b_v, \qquad ra_v + lb_v = 0, \tag{4}$$

For the representative $0 \le c < n$, define $R(c) = \#\{0 \le i < c : \tau_i = \mathrm{R}\}$ and $L(c) = \#\{0 \le i < c : \tau_i = \mathrm{L}\}$.

Proof. Choose an allowed heading at v and a coloring with $C(v) = c$. Write $C^{v,+}$ for the coloring equal to C away from v and with value $c + 1 \pmod n$ at v; let h' be the resulting heading. The update affects no other cell. Equation (2) gives

$$F(C^{v,+}) - F(C) = \kappa + \phi(v, h) - \phi(v + e_{h'}, h').$$

The right side is independent of all other colors. Increment the finitely many nonzero coordinates of C from zero, in any fixed order. Their increments therefore add, proving (3). The single-cell colorings fix the f_v uniquely. Equal-turn colors have the same target pose and hence the same increment: a_v for R and b_v for L. Summing these increments gives the first part of (4); going around the color cycle gives the second. □

The lemma does not say that every invariant of the ant is additive. The separation in (2) is an essential hypothesis. A head correction that also sees nearby colors is outside that hypothesis.

3 A complete face presentation

Write $d_v = a_v - b_v$. A square is an *R-face* when its lower-left corner has odd checker parity, and an *L-face* when that corner has even parity. Write ∂S for the four corner vertices of a unit square S. The directed four-pose circuit around an R-face uses four right turns; around an L-face it uses four left turns. For example the L-face based at $(0,0)$ has incoming poses $(0,0,\mathrm{S}), (1,0,\mathrm{E}), (1,1,\mathrm{N}), (0,1,\mathrm{W})$.

Theorem 3.1 (Face presentation)**.** *For a mixed cyclic rule, weights* (4) *admit a pose correction of constant rate κ in sector zero if and only if*

$$2d_v = 0, \qquad ra_v + lb_v = 0 \qquad \text{at every site}, \tag{5}$$

$$\sum_{v\in\partial S} a_v = 4\kappa \qquad \text{for every R-face } S, \tag{6}$$

$$\sum_{v\in\partial S} b_v = 4\kappa \qquad \text{for every L-face } S. \tag{7}$$

The pose correction is unique up to a constant.

Proof. At an even site, compare the N and S input equations with the same E or W target pose. The two comparisons yield

$$\phi(v,\mathrm{S}) - \phi(v,\mathrm{N}) = a_v - b_v = b_v - a_v.$$

Thus $2d_v = 0$. At an odd site the identical calculation gives $\phi(v,\mathrm{W}) - \phi(v,\mathrm{E}) = d_v$. The color-cycle relation was proved in Lemma 2.1. Summing the four pose equations on a face circuit gives (6) or (7).

For sufficiency let $U(v)$ be the correction at the primary heading, N on even sites and E on odd sites. The opposite heading is assigned $U(v) + d_v$. Orient horizontal grid edges from even sites to odd sites and vertical edges from odd sites to even sites. Assign the following increments of U:

Source parity	Edge to u	$U(u) - U(v)$
even	east	$\kappa - a_v$
even	west	$\kappa - a_v + d_v - d_u$
odd	north	$\kappa - a_v + d_v$
odd	south	$\kappa - a_v - d_u$

Reverse traversal negates the increment. Its circulation is $4\kappa - \sum a_v$ on an R-face, and $4\kappa - \sum b_v$ on an L-face; the latter simplification uses $2d_v = 0$. Both vanish.

Every finite closed walk in the square grid is a sum of elementary face boundaries and edge backtracks. This can also be proved by successively interchanging neighboring horizontal and vertical steps and removing backtracks. Consequently path integration from the origin defines U

on all of $\mathbb{Z}^2$. Substitute $\phi = U$ at primary headings and $\phi = U + d$ at opposite headings into the local transitions. The table gives the required equations; the remaining opposite-heading equalities use $2d = 0$. The color-cycle relation ensures that the functions f_v wrap consistently. This proves sufficiency. The edge gradients and the opposite-heading differences determine every pose once $U(0, 0)$ is fixed, proving uniqueness. □

This presentation includes arbitrary spatial coefficients and arbitrary abelian targets. The rule order still enters the actual color functions through $R(c), L(c)$; it disappears only from the equations on a, b. The latter reduction is the order-blindness mechanism identified in [3].

4 The maximal phase period

Theorem 4.1 (Sharp phase order)**.** *Let M be given by* (1)*. For any abelian group G, there is a separated cocycle of rate κ in one fixed sector if and only if $M\kappa = 0$. Its coloring coefficients and pose correction can be chosen four-periodic in each coordinate. If the same coloring function must work in both sectors, existence is equivalent to $4n\kappa = 0$.*

Proof. First suppose the rule is mixed. From (5),

$$na_v = ld_v, \qquad nb_v = -rd_v.$$

Multiplying a face equation by n gives

$$4n\kappa = l \sum_{\partial S_R} d_v = -r \sum_{\partial S_L} d_v, \tag{8}$$

where any one face of each kind may be used. Since $2d_v = 0$, $8n\kappa = 0$ is necessary. If l is even the first expression vanishes; if r is even the second vanishes. This gives the required necessary order.

For existence when $4n\kappa = 0$, write $z(x, y) = \mathbf{1}_{x \text{ odd}, y \text{ odd}}$ and choose

$$a_v = b_v = 4z(v)\kappa. \tag{9}$$

Every face contains exactly one odd/odd vertex. All site and face equations hold, so Theorem 3.1 applies.

Now suppose r, l are odd, so n is even, and $8n\kappa = 0$. Put

$$\begin{aligned} \chi(x, y) &= (-1)^{x+y}, \\ u(x, y) &= (1 - n/2)z(x, y) - (n/2)\chi(x, y)xy, \qquad (10) \\ a_v &= 4u(v)\kappa, \qquad b_v = (1 - n)a_v. \qquad (11) \end{aligned}$$

Then $d_v = na_v$ has $2d_v = 0$, and $ra_v + lb_v = n(1 - l)a_v = 0$, because $1 - l$ is even. The corner sum of χxy is the checker sign of the lower-left corner. It follows that the corner sum of u is 1 on R-faces and $1 - n$ on L-faces. Thus (6) holds and

$$\sum_{\partial S_L} b_v = 4(1 - n)^2\kappa = 4\kappa,$$

since $4n(n - 2)\kappa = 0$. Theorem 3.1 proves existence. Taking $\kappa = 1$ in $\mathbb{Z}/(8n)$ attains the upper bound.

The coefficients in (11) have period four: the change after $x \mapsto x + 4$ is a multiple of $8n\kappa$, and similarly for y. The edge increments in the proof of Theorem 3.1 are also four-periodic. Along

either coordinate axis their four-edge sum is zero (the terms alternate $\kappa, -\kappa$). Zero face circulation carries this identity to every parallel line. Hence the integrated U, and then ϕ, can be four-periodic. Construction (9) has the same property.

For a constant all-R or all-L rule, the blank ant circles four cells. After $4n$ updates its complete coloring and pose return, so $4n\kappa = 0$ is necessary for *any* state observable of constant increment. For sufficiency set $a_v = b_v = 4z(v)\kappa$, where $z(v)$ is the indicator that both coordinates are odd. Then $d_v = 0$, $na_v = 0$, and every unit square has corner sum 4κ. The edge table in the sufficiency proof of Theorem 3.1 therefore has zero circulation even in this constant-rule case. Its path integral U supplies $\phi(p, h) = U(p)$, and direct substitution verifies every allowed transition with $f_v(c) = 4c\, z(v)\kappa$. This constructs the clock whenever $4n\kappa = 0$, including a unit clock modulo $4n$.

Finally suppose one field function F works in both sectors for a mixed rule. Its increments a_v, b_v are common to the sectors. Every geometric face is an R-face in one sector and an L-face in the other. Therefore both corner sums equal 4κ, and every corner sum of d vanishes. Multiplying the sum of a by n gives $4n\kappa = 0$. Construction (9) uses one and the same F in both sectors and is sharp. □

Corollary 4.2 (Spatial resolution)**.** *If the increments a_v, b_v are two-periodic in either coordinate, a separated constant-rate cocycle necessarily satisfies $4n\kappa = 0$, even without a periodicity assumption on ϕ. When r, l are odd, four-periodic coefficients attain order $8n$.*

Proof. Under two-periodicity in x, a four-corner sum is unchanged by moving the face one column. The neighboring face has the opposite type. Hence both sums, for a and b, equal 4κ on every face. The final argument in Theorem 4.1 applies. The y case is identical. □

Thus the small sublattice ansatz is a genuine restriction for detecting the extra phase. For example RL has maximum 8 under that restriction and 16 without it; RRRL has 16 and 32, whereas RRLL has 16 in both cases.

5 The classical ant: a sixteen-phase formula

For the rule RL, write B for the complete finite black set. The two increments are $a_v = w_v$, $b_v = -w_v$, and $4w_v = 0$.

Theorem 5.1 (Complete classical separated form)**.** *In sector zero, an observable* (2) *exists precisely when $16\kappa = 0$ and*

$$F(B) = F(\varnothing) + \sum_{(x,y)\in B} (-1)^{x+y}\big(f(x) + g(y) - 4xy\kappa\big), \tag{12}$$

where f, g take values in $G[4] = \{a : 4a = 0\}$. One may normalize $g(0) = 0$; then the functions are unique. The pose correction is unique up to a constant.

Proof. Lemma 2.1 first forces F to be a sum of weights w_v. For a square based at (x, y) the two face equations reduce to

$$\sum_{v\in\partial S} w_v = -4(-1)^{x+y}\kappa.$$

Let $A(x, y) = (-1)^{x+y}w(x, y)$. Its mixed difference is -4κ. Telescoping over rectangles, including signed rectangles for negative coordinates, gives

$$A(x, y) = A(x, 0) + A(0, y) - A(0, 0) - 4xy\kappa.$$

Take $f(x) = A(x,0)$ and $g(y) = A(0,y) - A(0,0)$. Both are 4-torsion. Applying 4 to the mixed difference forces $16\kappa = 0$. Conversely, these conditions make every weight 4-torsion and satisfy both face equations, so Theorem 3.1 applies. □

At zero rate this recovers the classical checker-signed row and column charges; the underlying local cocycle condition is due to [2]. The theorem adds the nonzero-rate term, sharp order, and the removal of an a priori additivity assumption.

For an explicit unit clock modulo 16 take $f = g = 0$. Set

$$U(x,y) = \begin{cases} 0, & x,y \text{ even}, \\ 1, & x \text{ odd},\ y \text{ even}, \\ 4y-2, & x,y \text{ odd}, \\ 4x-1, & x \text{ even},\ y \text{ odd}, \end{cases} \qquad \Phi(x,y,h) = U(x,y) + 8xy\mathbf{1}_{h\in\{\mathrm{S,W}\}}. \tag{13}$$

Then

$$Q(B,p,h) = -4 \sum_{(x,y)\in B} (-1)^{x+y} xy + \Phi(p,h) \pmod{16} \tag{14}$$

satisfies $Q(T_{\mathrm{RL}}X) = Q(X)+1$. This follows either by substitution in the eight heading/color cases or from the integrated construction above. For sector one evaluate the formula on the quarter-rotated state. In particular,

$$t \equiv Q(X_t) - Q(X_0) \pmod{16}. \tag{15}$$

All black cells enter the finite sum, including cells not yet read. The formula is spatially four-periodic, and its evaluation needs no past trajectory.

Corollary 5.2. *No separated unit clock modulo 32 exists for classical RL on one full sector, even with an arbitrary nonlinear F and arbitrary spatial pose correction. For any mixed cyclic rule there is no nonconstant separated first integral with values in a torsion-free abelian group, and no such clock of a fixed positive real increment.*

Proof. The first assertion is Theorem 4.1. For a torsion-free target, that theorem forces $\kappa = 0$. Equations (5) give $d = 0$ and $na = 0$, hence $a = b = 0$. The integrated pose correction is constant. □

This does not exclude strict clocks whose increment is positive but varies with the state. It also does not put a bound of sixteen on information available to a genuinely joint function of the complete state.

6 All compact separated invariants

We now set $\kappa = 0$ and assume the rule is mixed. Call F *compact* when it depends only on the colors in some fixed finite set of sites. By Lemma 2.1, this means that a, b are finitely supported.

Theorem 6.1 (Compact-charge basis)**.** *Modulo additive constants, the compact separated invariants over G form*

$$\left(\bigoplus_{S \text{ an L-face}} G[2] \right)_{r \text{ even}} \oplus \left(\bigoplus_{S \text{ an R-face}} G[2] \right)_{l \text{ even}}, \tag{16}$$

where a parenthesized summand is omitted when its subscript condition fails. Every invariant is a unique finite sum of the plaquette generators below. Thus classical RL has no nontrivial compact separated field invariant, over any abelian group.

Proof. Since $2d = 0$, we have $2a = 2b$. Doubling the two face conditions shows that $2a$ has zero corner sum on every square. Therefore $\chi\, 2a$ is a sum of a row function and a column function. A finitely supported function of this form is zero: outside its horizontal support the column function must be constant, and outside its vertical support the row function must cancel that constant. Hence $2a = 2b = 0$.

The color-cycle equation is now $(r \bmod 2)a + (l \bmod 2)b = 0$. If both counts are odd, $a = b$ and the all-face zero-sum argument forces $a = b = 0$. If r is even and l odd, $b = 0$ and a has only its R-face constraint. The other case is symmetric. If both counts are even, the R-face constraint on a and the L-face constraint on b are independent.

To solve the remaining R-face constraint, make a graph whose vertices are the centers of the R-faces. Each original lattice vertex joins its two diagonally opposite incident R-faces, and is an edge of this rotated square lattice. The condition $\sum_{\partial S_R} a = 0$ says exactly that this finitely supported edge assignment is a cycle with coefficients in $G[2]$. Every such cycle is a unique finite sum of elementary face boundaries. One elementary proof is to integrate edge labels along dual paths starting outside a bounding box of the support, assigning exterior faces coefficient zero. Zero boundary makes the integration path independent; finite support leaves only finitely many faces with nonzero coefficients. If every edge difference is zero, all face coefficients equal the exterior zero, proving uniqueness. Elementary faces of this graph correspond to the original L-faces. This proves the first direct sum in (16); exchange the face colors for the second.

It remains to provide the actual pose corrections. For an L-face S, let $\Pi_L(S)$ be the four incoming poses of its all-L circuit. If r is even, and $z \in G[2]$, define

$$J^R_{S,z}(C, p, h) = z \left(\sum_{v \in \partial S} R(c_v) + \mathbf{1}_{(p,h) \in \Pi_L(S)} \right). \tag{17}$$

The wrap in color is well defined because r is even. A right turn at a corner changes the field sum by z and moves the pose into or out of $\Pi_L(S)$, changing its correction by z; all other transitions change neither contribution. Thus the total is conserved. When l is even the analogous J^L uses an R-face, $L(c)$ and the all-R pose circuit. Their coefficients are exactly the elementary boundaries already classified, so they realize and exhaust the direct sums, with unique normalized pose corrections. □

The condition is on the total counts, not on consecutive run lengths. For example RLRL has both families in (16), despite its odd run lengths. This is distinct from the even-run hypothesis used for recurrent symmetry in [1]; no conclusion about recurrence follows here.

7 Verification and reproducibility

The proofs above are analytic over all parameters and all abelian coefficient groups. The accompanying implementation performs three kinds of exact control. First, it checks every color transition at every four-periodic coordinate and allowed heading for all rules of lengths one through eight, including color wrap. It also evaluates local identities at very large integer coordinates. Second, an independent raw linear presentation assigns a separate unknown to every site/color and every pose, without assuming same-turn reduction. Exact integer Smith forms on two- and four-periodic grids recover the respective phase orders. Finite-box binary kernels and integer presentations recover the compact basis and its order-two factors. These finite presentations do not replace the all-plane proof.

Third, independent Python and C++ literal dynamics compare complete read-color streams and complete final colorings for finite inputs in both sectors, including unread remote cells. The Python

head potential is computed by coordinate-path integration; C++ reconstructs its potential independently on the raw pose graph. Both periodically recompute the entire coloring contribution. The classical closed formula (14) is checked separately. The deterministic report records performed counts, negative controls and source hashes. Reproduction uses `python phase_cocycle/reproduce.py`; exact matrix checks require SymPy and the native control requires a C++17 compiler.

8 Limitations

The maximality theorem concerns the separated, constant-rate, all-state class (2). It does not classify arbitrary joint-state functions, pose corrections that inspect colors, variable-increment potentials, or identities restricted to a selected orbit. A modular clock does not decrease and cannot by itself be a finite-resource termination argument. Neither the phase classification nor the compact-charge basis proves eventual periodicity, period-104 uniqueness, the finite-support Highway Conjecture, or universality. The new classification is not a claim that the earlier classical charges or same-turn reduction were unknown. The verification is unformalized analytic proof with exact computational controls, not a theorem-prover formalization.

Computational assistance. Mathematical exploration, source drafting and test implementation were assisted by a language model. All claimed general results have explicit proofs above; finite tests are reported only for their stated verification scope.

References

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